

Future technologies, dystopic futures and the precautionary principle

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Abstract. It is sometimes suggested that new research in such areas as artificial intelligence, nanotechnology and genetic engineering should be halted or otherwise restricted because of concerns about possible catastrophic scenarios. Proponents of such restrictions typically invoke the precautionary principle, understood as a tool of policy formulation, as part of their case. Here I examine the application of the precautionary principle to possible catastrophic scenarios. I argue, along with Sunstein (*Risk and Reason: Safety, Law and the Environment*. Cambridge University Press, Cambridge, 2002) and Manson (*Environmental Ethics*, 24: 263–274, 2002), that variants of the precautionary principle that appear strong enough to support significant restrictions on future technologies actually lead to contradictory policy recommendations. Weaker versions of the precautionary principle, which do not have this feature, do not appear strong enough to support restrictions on future technologies.

Key words: catastrophe, cost–benefit analysis, grey goo problem, nanotechnology, precautionary principle

Introduction

The prospect of far reaching developments in artificial intelligence, nanotechnology and genetic engineering have prompted the expression of fears about possible catastrophic scenarios. For example, Drexler (1986) argues that developments in nanotechnology may lead to the creation of self-assemblers that will be able to convert everything in the universe into themselves. Eventually the universe will be composed only of self-assemblers. This is known as the “grey goo” problem (Drexler 1986: 172). It has also been suggested that developments in artificial intelligence have the potential to lead to the creation of robots or post-humans that will be vastly more intelligent than humans and that these super-intelligent beings may have the ability and the desire to exterminate humans. A third catastrophic scenario that many have worried about is the possible use of electronic surveillance technology to erode individual privacy and perhaps to create an Orwellian surveillance society.¹

¹ Concerns about the possibility of a surveillance society are almost always associated with Orwell’s (1984). However, similar concerns have been raised by many authors. See Lyon (1994), especially Chapter 4. The Surveillance Camera Players are an activist group that seek to raise awareness about the potential risks of video surveillance. See <http://www.notbored.org/the-scp.html>.

It is extremely hard to say exactly how likely it is that such dystopic possibilities will occur, even if it seems safe to say that they are most unlikely. It is also extremely hard to know how to assess the costs of such possibilities, even if it seems safe to say that they are astronomically high. How are we to factor these catastrophic possibilities into policy-making? Here we can distinguish at least three approaches. A first approach is to ignore such possibilities and only address risks that we can quantify, when making policy regarding future technology. A second approach is to attempt to assess the likelihood of such possibilities and the costs of such dystopic scenarios. A third approach is to ignore other considerations, in the face of such possibilities and, taking the attitude “better safe than sorry”, institute a moratorium, or some other form of restriction, on research that might lead to such dystopic scenarios. This last approach has been suggested by the ETC group, in respect of nanotechnology research. The ETC group, a Canadian-based lobby group “... dedicated to the conservation and sustainable advancement of cultural and ecological diversity and human rights” (<http://www.etcgroup.org/about.asp>), call for an immediate moratorium on nanotechnology research and the production of new nanomaterials, “... until such time as laboratory protocols and regulatory regimes are in place to protect workers and consumers, and until these materials are shown

to be safe" (ETC Group 2005: 16). They further demand that currently available products that incorporate manufactured nanoparticles be withdrawn from shop shelves (ETC Group 2005: 17).

Applications of either of the first two approaches sketched above are probably best understood as instances of cost–benefit analysis. In applying cost–benefit analysis we make policy recommendations on the basis of a calculation of the relevant possible benefits and possible costs of particular outcomes and then compare our results with cost–benefit analyses for other possible outcomes. The third approach, however, is not easy to interpret as an instance of cost–benefit analysis, because the possible benefits of a policy are entirely excluded from consideration. It is probably best understood as an application of the "precautionary principle" a principle of policy formulation that has been developed largely in the context of environmental law, but which is now being applied in a variety of contexts in policy debates and in the regulation of new technologies.²

Defining the precautionary principle

New technology can and does lead to a variety of undesirable and unexpected consequences. Telephone systems crash, data base errors lead to over and under-billing by government agencies and bugs in software systems lead to accidents as well as deaths (Baase 1997: 114–129). With the benefit of hindsight it is easy to see many circumstances in which it would have been better if we had been more cautious. Merely to argue for greater caution is not yet to advocate the precautionary principle. Advocates of the precautionary principle do not simply argue that we are liable to underestimate risks. Rather, they argue that we should not act on the basis of risk assessments, as such. Instead, it is argued that we should always err on the side of caution. It is, say the proponents of the precautionary principle, "better to be safe than sorry".

The fact that there is a definite description in common usage – "the precautionary principle" – might seem to suggest that there is an agreed upon formulation of this principle. Nothing could be further from the truth. The precautionary principle is

variously defined and its critics are wont to complain about the lack of a clear definition of the principle (Bodansky 1991) and of its lack of apparent logical structure (Gray and Bewers 1996: 768). What different versions of the precautionary principle have in common is that they stress that we should be willing to act to avoid bringing about a possible set of undesirable circumstances, even when we lack firm evidence to suggest that such an outcome is even possible.

One well-known and influential formulation of the precautionary principle is the Wingspread Statement of 1998:

Where an activity raises threats of harm to the environment or human health, precautionary measures should be taken even if some cause and effect relationships are not fully established scientifically (Wingspread 1998).

Another influential formulation, developed in the context of environmental law, is Principle 15 of the 1992 *Rio Declaration on Environment and Development*:

In order to protect the environment, the precautionary approach shall be widely applied by States according to their capabilities. Where there are threats of serious or irreversible damage, lack of full scientific certainty shall not be used as a reason for postponing cost-effective measures to prevent environmental degradation (United Nations Environment Programme 1992).

Neil Manson suggests that formulations of the precautionary principle, such as these, share a common structure as follows:

If an e-activity is {choose one or more damage conditions} and if it is {choose knowledge condition} that the e-activity causes the e-effect, then decision makers are obliged to {choose one or more e-remedies} (Manson 2002: 267).

"e", in this context, refers to the environment, broadly understood.

Not all versions of the precautionary principle can be fitted under this framework, however, consider the following definition:

Where there are significant risks of damage to the public health, we should be prepared to take action to diminish those risks, even when the scientific knowledge is not conclusive, if the balance of likely costs and benefits justifies it (Horton 1998: 252).

On this version, the precautionary principle no longer appears to be a tool of policy formulation that is distinct from cost–benefit analysis. Instead it appears to be a way of framing cost–benefit analysis so as to ensure that those who use it do not forget

² For example, the "Stewart Committee Report" (Independent Expert Group on Mobile Phones 2000) argues for the application of the precautionary principle in relation to the use of mobile phones. Also the European Group on Ethics (2005) has recently argued for the application of the precautionary principle to the use of ICT implants in the human body.

that scientifically inconclusive risks are still risks that need to be considered when making policy.

Should attempted statements of the precautionary principle that can be understood as variants of cost-benefit analysis be considered to be genuine attempts to define the precautionary principle? In so far as they capture the sentiment of those who promote the attitude "better safe than sorry", they share a significant common feature with other variants of the precautionary principle, so it seems that they should be so counted. However, the existence of such weakened versions of the precautionary principle raises the suspicion that, when placed under argumentative pressure, proponents of the precautionary principle who might prefer to be advocates of a distinct policy tool, tend to end up advocating a policy tool that is simply a variant of a conventional approach to policy formulation.

It seems that under the first two versions of the precautionary principle that we have considered, the precautionary principle could be used to mount a straightforward argument for a moratorium, or at least for the placement of severe restrictions, on nanotechnology research, in light of the grey goo problem. However, on the third version, it would seem to be very hard to justify a moratorium, or such restrictions. Once the possible benefits of research and the low probability of the grey goo problem occurring are taken into account, the influence of this low probability catastrophic scenario on policy is correspondingly lessened.

The precautionary paradox

Manson (2002: 270–274) and Sunstein (2002: 102–104) both argue that stronger versions of the precautionary principle entail a paradox and lead to contradictory policy recommendations. To appreciate the paradox, consider again the grey goo problem. It is possible, as Drexler (1986: 172–173) argues, that our efforts to conduct research in nanotechnology will cause the grey goo problem. However, it is also possible that our failure to conduct research in nanotechnology will lead indirectly to the grey goo problem, or some other catastrophic scenario. For example, it seems possible that research in nanotechnology that we did not conduct might have led to us finding a way to prevent the grey goo problem and it seems possible that on another planet in the universe a civilization flourishes that conducts research into nanotechnology that leads to the creation of self-assemblers that begin to convert everything into themselves. If we fail to conduct research that leads us to find a way to prevent the grey goo problem and if the alien civilization in our example does conduct research that leads to the creation of self-assemblers that begin to convert everything into

themselves, then our failure to conduct research in nanotechnology will have contributed to causing a catastrophe, because we will be unable to prevent the grey goo problem. So the precautionary principle leads us to conclude both that we should conduct research into nanotechnology and that we should not conduct research into nanotechnology.

It may seem tempting to dismiss the above scenario as so unlikely as to be irrelevant, or as even less likely than the catastrophic scenario which prompted our hypothetical application of the precautionary principle, in response to the grey goo problem. However, neither of these responses seem available to the proponent of the application of strong versions of the precautionary principle to catastrophe scenarios. Both responses involve assessing the relative probabilities of different possible future scenarios, and these are exactly the sort of calculations that we are told to forego by proponents of strong versions of the precautionary principle. We might, of course, retreat to a weak version of the precautionary principle, but as I have already argued, this is to retreat all the way to a version of cost-benefit analysis, so we would no longer be applying an interesting and distinct principle of risk analysis in the service of policy formulation.

I am aware of three possible responses to the charge that the precautionary principle leads to contradictory policy recommendations. The first is that we should not apply the precautionary principle when there is no evidence of a potential hazard (Sandin et al. 2002: 291). The second is that we should apply the precautionary principle to itself, before attempting to apply it further (Sandin et al. 2002: 294). The third is that we should construe the precautionary principle as allocating the burden of proof (Saunders and Ho 2000). I will consider each in turn.

Sandin et al. (2002) claim that Manson and other critics misunderstand the precautionary principle. According to Sandin et al. (2002: 291):

The principle requires that actions be taken when there is a lack of full scientific certainty. This, however, does not mean that precautionary measures are required when there is no particular evidence, scientific or other, of the presence of a possible hazard.

The problem with this line of defence is that it is exceedingly unclear how to go about differentiating between situations where there is less than full scientific certainty and situations where there is no evidence of a possible hazard.³ Into which category are we to

³ Majone (2002: 104) makes a similar complaint in the context of criticism of the European Commission's Communication on the Precautionary Principle.

place the grey goo problem, for example? Is there evidence of its possibility? There are currently no self-assemblers, so one might argue that there is no evidence of its possibility. However, the grey goo problem appears to be a logically coherent scenario and technological advances in nanotechnology make it seem to be a realisable scenario. Indeed, there are attempts to specify solutions to the grey goo problem, surely a good indication of its possibility.⁴ The defence of the precautionary principle that Sandin et al. (2002) are trying to run requires that we find there is no evidence of catastrophic scenarios such as the grey goo problem. However, they do not do nearly enough to specify what would count as acceptable evidence to be able to rule that such possibilities lack a suitable evidential basis.

Sensing that their defence of the precautionary principle against the charge of providing contradictory policy recommendations will be judged unsatisfactory because it trades on such a vague distinction, Sandin et al. (2002: 291) suggest that the defence can be recast as a recommendation of the application of a *de minimis* rule. In effect this means that we only apply the precautionary principle when we have identified risks that are over a certain threshold of probability. This line of defence of the precautionary principle risks being a toothless one, depending on where the *de minimis* threshold is set. If the threshold is set too high, then the precautionary principle will not be applicable to many of the risks that its advocates wish to apply it to. On the other hand, if the threshold is set too low then the problem of paradox won't be avoided, because the precautionary principle will continue to issue contradictory advice.

A second suggestion, also due to Sandin et al. (2002), is that in order to ensure that countervailing risks are not overlooked when we attempt to apply the precautionary principle, that "the precautionary principle should be applied also to the precautionary measures prescribed by the precautionary principle itself" (2002: 294). While this suggestion has the virtue of even-handedness in its favour, it is disastrous for defenders of the precautionary principle, because it threatens to leave us in a situation where we are almost never able to apply the precautionary principle to any real life situation, as it is almost always possible that the act of taking precautionary measures will lead to the possibility of some or other catastrophic risk arising.

Saunders and Ho (2000) consider the charge that the precautionary principle leads to contradictory outcomes, in the context of debates about genetically

modified foods, and they hold that the way of defending the precautionary principle from this charge is to argue that it should not be interpreted as a rule for decision making, but as a rule for allocating the burden of proof. The burden of proof falls on proponents of genetically modified foods to justify their research, in light of the fact that the evidence of threats of harms ensuing from genetically modified foods outweigh the risks of failing to conduct such research, or so Saunders and Ho (2000) appear to argue.

One's immediate response to this line of defence is that it reduces the precautionary principle to a mere postscript to cost-benefit analysis. Saunders and Ho (2000) might usefully recast their argument as the claim that if cost-benefit analysis leads us to the conclusion that we should not risk growing genetically modified foods, then the onus is on proponents of genetically modified foods to find additional evidence that might make us change our minds. If this is right, then it would seem that cost-benefit analysis must be conducted, to determine how to apply the precautionary principle. A second reaction is that the defence appears somewhat disingenuous. Saunders and Ho (2000) make the claim that critics are wrong to interpret the precautionary principle as a principle for decision making. However, the majority of both the precautionary principle's defenders and critics appear to construe it as just such a principle.

Given that strong versions of the precautionary principle fail to prescribe definite courses of action, in many of the circumstances that people wish to apply the precautionary principle, it seems that we need to explain why it is that so many people believe that the precautionary principle does prescribe definite courses of action in many circumstances. Sunstein (2003) attempts to address this question by drawing on the resources of behavioural economics and cognitive psychology. He considers a number of cognitive biases which have the potential to distort our reasoning about risk, leading us to overestimate the importance of some risks and to be oblivious to other risks (2003: 1035–1054). For example, the availability heuristic ensures that certain risks that are already familiar to us are present to consciousness when we are considering the possible consequences of our actions. If we are familiar with the events of Three-Mile Island and/or Chernobyl, then the availability heuristic operates to ensure that the risks of accidents are present to consciousness when we consider the pros and cons of operating nuclear power plants. If the risk of operating nuclear power plants seem particularly salient to us, then the effect of such cognitive biases may be to distract us from even

⁴ See, for example, Freitas (2001).

considering the risks of not operating nuclear power plants (such as the risks associated with climate change, due to the increased use of alternative sources of fuel). Rationally, we should consider both the risks associated with operating nuclear power plants and the risks of not operating nuclear power plants before deciding whether or not to operate nuclear power plants. However, humans often do not think this way. Instead we tend to focus on a particularly salient possible outcome and fail to consider other possibilities. The widespread belief that the application of strong versions of the precautionary principle lead to particular policy outcomes appears to be a consequence of such systematic biases in human reasoning.

Vagueness as a virtue

While many will see the failure of proponents of the precautionary principle to agree on a working definition of the principle to be a shortcoming, Jordan and O'Riordan (1999) see the vagueness of the principle as a virtue. According to them:

... we conclude that the application of precaution will remain politically potent so long as it continues to be tantalizingly ill-defined and imperfectly translatable into codes of conduct, while capturing the emotions of misgiving and guilt (15).

They seem to see the precautionary principle not so much as something to apply to particular cases, in order to create policy, but rather as a guide to influencing the way policy makers understand the world. In essence, for them, advocacy of the precautionary principle is a way to help ensure greater caution and increased concern for the environment and for the social consequences of technology. Jordan and O'Riordan (1999) highlight seven themes that the precautionary principle embodies. In summary, these are:

1. *Pro-action* – Preventative measures should be taken even where there is lack of scientific evidence of any possible harm resulting from a course of action;
2. *Cost-effectiveness of action* – There should be examination of “identifiable social and environmental gains” resulting from a course of action that might justify overall costs;
3. *Safeguarding ecological space* – When considering the environmental impacts of a course of action we should always allow a wide margin of error. The environment should not be pushed to the limit;
4. *Intrinsic value and legitimate status* – Vulnerable natural systems should be protected “as a matter of moral right”;

5. *Shifting the onus of proof* – the burden of proof should be placed on those who favour development to show that it is safe;
6. *Meso-term planning* – More emphasis in policy making should be placed on “future and uncertain issues”;
7. *Paying for ecological debt* – Those who have not been suitably cautious in the past bear responsibility for the consequences of their actions, even if these were undertaken in ignorance.

(Jordan and O'Riordan 1999: 24–29)

Jordan and O'Riordan (1999) tell us something enlightening, I think, about the precautionary principle. Advocates of the principle may have aimed to develop an alternative to conventional cost-benefit analysis. However, their underlying concern has been with the narrow scope of the considerations that policy makers, who have utilised cost-benefit analysis, have usually included in their calculations. In particular, advocates of the precautionary principle have actually been concerned that considerations that did not have the backing of scientific certainty were not being taken into account by policy makers, and they were actually concerned that the broad social and environmental costs of particular policies were not being considered. In trying to develop an alternative to cost-benefit analysis, however, it seems that advocates of the precautionary principle have aimed at the wrong target. The problem that really concerned them was not with cost-benefit analysis itself, but with the ways in which it has been employed.

As Jordan and O'Riordan (1999) argue, the precautionary principle has functioned effectively as an intellectual tool of political protest movements, in part because it has been vaguely formulated. However, it cannot reasonably be expected that the precautionary principle will achieve the status of a principle of mainstream policy development without more precise formulation. Once the principle is used to attempt to implement policy it becomes crucial that its implementers are able to agree on the consequences that follow from any particular attempt to apply the policy. So, the vagueness defence of the precautionary principle fails, if it is understood as a defence of the principle as a tool to be used when implementing policy, even if it succeeds as a defence of the precautionary principle as a tool of political protest.

Concluding remarks

Because the precautionary principle has been vaguely formulated, it has been possible for some of its

proponents to employ it as a component of an argument for the conclusion that we should ban or restrict research into certain areas of technological development, in virtue of the possibility of a catastrophic scenario being realised. Once we gain a better understanding of the precautionary principle, we can see that these particular arguments are unsustainable. Versions of the precautionary principle that may appear strong enough to support such responses actually lead to contradictory policy recommendations. Versions of the precautionary principle that do not lead to such contradictory policy recommendations are not strong enough to support such extreme policy recommendations.⁵

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